Vision statement

Freek Pols

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by

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1. Introduction

Dear committee member,

It is an honour and a pleasure to share my vision statement on my career at Delft University of Technology. Reflecting on my 20^+ years in education as a teacher and innovator, my performance over the past seven years at TU Delft, and my aspirations for growth as an educator, scientist, and person has been an insightful journey.

The summary of my vision can be expressed as:

> *To strive for continuous growth and excellence in education, bringing out the best in both myself and others.*

But even with this clear, concise and strong statement - covering one of my core values, it has been a challenge to draft a clear document in which education, research, and valorization are distinguishable components. I started as a secondary school teacher, did alongside teaching a Ph.D. study in physics education and continued my career in academia with a focus on education. As a result, my research and teaching are inherently intertwined: I study what I teach, and how I teach is grounded in research (scholarly-teaching). With a drive to openly share all educational material (hence, contributing to open education), the creation of evidence-based education (education & research) and valorization (enabling life long learning) are for me inseparable, though I have attempted to distinguish them in this document.

This document, written in MyST - also known as JupyterBook - exemplifies my innovative approach to teaching: it is one of the latest technological advancements in education and research dissemination [1]. The MyST and Jupyter projects aim to create interactive scientific publications for the web, with the ability to export content to PDF, LaTeX, and Microsoft Word. While this document is presented as a PDF, it is also available as an interactive online document. Moreover, this technology is used for the latest online interactive textbook publications, see for instance TUD's first year physics mechanics book. Where in the past few years I have foremost used this technology (as 'consumer'), I now have come to a stage where I actively contribute to the development of the MyST ecosystem (as 'producer'). For instance, this PDF version is made with a self-made Typst Thesis Template which is now open access available and part of the Jupyter Book ecosystem. It is a template made especially for students writing their bachelor's or master's thesis. ~~How the engagement with such new technologies in education changes how we teach, and how it improves the quality of our teaching is elaborated on throughout this document.~~ But it has been designed in such way that it can be extended to several other purposes

x ref to dissertation

1.1 How to read this document

Chapter 2 presents evidence of my work as an educational innovator who designs, coordinates, and continuously improves education. The focus is not on isolated teaching activities, but on systematic transformation of courses, curricula, and teaching practices.

Chapter 3 presents my work as a physics education researcher with a focus on integrating argumentation in both scientific inquiry and engineering design. Although my research is closely connected to my teaching, this chapter should be read as a contribution to the international research community.

Chapter 4 highlights my efforts in (especially) open education at local, national, and international levels.

Chapter 5 presents my vision on leadership related to teaching and innovating education and educational research at TU Delft.

The appendices include detailed descriptions of courses taught, research publications, and valorization activities. While these chapters contain important information, they should be read as supplementary material that substantiates the main narrative.

Sincerely yours,

2. Education

I think one of my first experiences with teaching is instructing sailing at an age of 14. At upper secondary school, I was a tutor for several years, helping fellow students with mathematics and physics. When I was 16, I became an instructor in a martial art (Pencak Silat). In hindsight, it thus seemed a logical step to do the minor education in my third year of the study Applied Physics and to become a secondary school teacher after my masters. I have been teaching physics in secondary school for ten years (2009-2019).

Recognizing that I probably had not yet reached my full potential and wishing to set an example for my pupils that learning is a lifelong journey, I applied for the NWO Doctoral Grant for Teachers. This grant was awarded in 2014, allowing me to do a PhD. study (0.4 FTE - 5 years) on teaching scientific inquiry in physics [2].

In 2019 I switched from secondary education to university and became the coordinator of the First Year Physics Lab Course (FYPLC) with the task to innovate this course.

2.1 Innovating lab courses

In 2019 I started my position at TU Delft as a lecturer/innovator (docent 2) of two lab courses: the *First Year Physics Lab Course (FYPLC)* and *Introduction to Experiments in Physics*, a course for the minor modern physics.

The first-year physics lab course (TN1405, 6 ECTS) was largely unchanged from its format in 2004 when I took the course as a first-year student. The course had run more or less the same for 40 years and was not much appreciated in the last two decades.

The course that was taught to third year minors (TN2985 - 2 ECTS) was an almost one-on-one translation of the first year course and was not much appreciated by the students.

The assignment given by Prof. Chris Kleijn (then program director) had three objectives:

- Modernize the courses to align with 21^{st}-century educational standards.
- Increase student motivation for the practicum.
- Focus on teaching students how to conduct research.

After a year of getting acquainted with the two courses, identifying the bottlenecks and possibilities (also for the administration that was still done on paper and therefore labor intensive), I started the innovation process in 2020. The results of the innovation have been published in several educational journals and conference proceedings - especially in C. F. J. Pols and P. J. J. M. Dekkers [3].

To summarize, the main changes in the FYPLC that were implemented are:

- Redesign of the entire course to focus on research skills and scientific thinking instead of following instructions to perform experiments.
- Introduction of a new practicum on determining g with 0.1% accuracy.
- Introduction of a final project where students design and conduct their own experiments (given some constraints).
- Introduction of a final exam covering Python, data analysis, measurement and uncertainty.
- Implementation of blended learning, reducing contact hours by 50% and allowing students to prepare at home.
- Digitization and automation of administrative tasks, enabling TAs to manage attendance and grading independently.
- Complete revision of the labcourse manual (twice), with links to lectures, assignments, interactive quizes and activities etc. [ref] we are frontrunner, based on this work other unis followed.
- Revision of grading rubrics.

A new program was developed for the minor course. Students now enjoy significant flexibility, both in progressing at their own pace (for Python and report writing) and in choosing experiments. As a result, student satisfaction increased from 5.7 (2019) to 8.0 (2020) to 8.2 (2021) and 8.4 (2025).

and a parallel course with 30-50 students doing two exp. in 2 weeks

Note that a 6-ECTS practicum with around 240 students cannot function without TAs. Major changes have been implemented here as well: Active recruitment has been established based on performance, engagement, and interest during the academic year. Efforts have also been made to retain TAs longer, partly through specialized ITAV training and targeted professional development, but also by granting them greater agency and responsibility, and by creating a supportive and inclusive TA environment. The total number of TA hours was reduced by scaling up, conducting the same experiments in a studio classroom, and designating one day for centralized (online) instruction (resulting in a yearly cost reduction of ∼30k€).

~25%

2.1.1 CI/CD

With the renewed physics curriculum of 2025, the FYPLC has been further adapted. Rather than one lab course, it now consist of two courses of 5 ECTS each, one in the first quarter and one in the second quarter. The first course was directly adopted from the previous FYPLC, no adjustments were needed. The second course, however, was redesigned to better align with the content of the second quarter (thermodynamics). Together with Roel Smit we designed nine laboratory activities and ten thermodynamic simulations aiming to develop better understanding of thermodynamic concepts at the macroscopic level through a focus on microscopic particle behavior, as seen in Figure 2.

everywhere

Figure 2.1: One of the simulations students have to develop where they compare Brownian motion of gas particles (red and blue) to a single heavier atom (black dot). The trajectories of a single light and the heavy particles are shown. Note that the full animation can be seen online only.

The development of this second course is still ongoing. Based on the first iteration, we have identified several pitfalls and areas for improvement. For instance, we noted students' reliance on genAI necessitating us to rethink how we provide and assess our labs. We currently transform some of the lab descriptions to a format that I developed [G. Pols [4]; F. Pols [5]][1]. The *scientific graphic organizer* can be regarded as a prestructured but simplified lab journal where all essential information is provided to

[1]The SGO, originally designed for physics labs is now used throughout the Netherlands in physics, biology and chemistry. The AAPT publication [5] was in the top 5 read articles of 2024, and we thus can assume it has an impact beyond the Netherlands.

meerdan alleen assessment

produce a fair judgement of students' doing and learning in and from lab work, see Figure 3. Without a computer and access to ChatGPT students now have to consider themselves what they are doing and why in that particular way. Moreover, we hope that this way of assessing students' understanding reduces the assessment load.

Figure 2.2: Some of the current labs are formatted in the SGO structure, a two page prestructured lab journal [5]. Here only the front is shown.

Another key aspect we aim to further develop is the use of version control systems (e.g. Git). The importance of coding, simulations, and collaborative workflows in physics has increased significantly over the past decades. However, many researchers have not been formally trained in these practices, making it difficult for students to grasp them when they are most needed—during bachelor's and master's projects.

the main ideas & workflows

Initial steps have been taken to introduce students to version control in this course, as addressing this early in their academic careers can help prevent common issues (challenges related to version control, data management, and maintaining an efficient and reproducible workflow) encountered later on. These aspects directly contribute to an essential component of our university's open science vision - which requires more explicit and structured integration within our educational programs. However, getting familiar with git is daunting and strategies to teach this effectively are limited. In the next two years we will further explore a feasible way to introduce ~~this~~ further to students. An interesting observation is that our introduction of git in Q2 has already let to revised approaches in subsequent courses.

Link

git workflows

2.2 Teaching in the teacher education program

I have been involved in the teacher-education program at TU Delft already for a long time. It started with teaching specialized topics as *labwork in science education* and *the use of ICT in physics education*. Over the years my involvement has increased, contributing (e.g.) to the development of a new course (implementation of education) - where students ought to consider the merits and trade-offs of different educational approaches (like doing an educational escaperoom) and test it in their own class.

With the upcoming retirement of the main physics teacher educator, my involvement in the program has increased, e.g. in supervising master thesis projects. Related to this is my senior university teaching qualification (SUTQ) project. The SUTQ project focused on *The development of a learning pathway on*

further

my engagement in the program

scientific inquiry in the teacher-education program. The teacher education program lacked a coherent teaching sequence on educational research whilst students of the two year masters program were still expected to do a research project. Moreover, the various teachers involved seem to have different opinions about the role and nature of educational research in the program. In the SUTQ project (2025-2026) I addressed *why, what, when, by whom, and how we are teaching educational research in TU Delft's teacher-education program.* This project has led to a blueprint for a learning pathway which is currently further developed and implemented. A paper has been submitted to [ref]

2.3 An educationalist with (pedagogical) content knowledge

At our university, we have many educationalists (onderwijskundigen). They are a great asset in many ways as they possess strong knowledge of education in general and can therefore support educators in designing their courses. However, when content-specific questions arise, this becomes more challenging as they educationalists often do not have in-depth knowledge of, for example, quantum mechanics, coding, or fluid mechanics - commonly referred to as *content knowledge.* While they may know in general what an assessment should look like, operationalizing this within a specific context, such as a Python course, can be difficult. Moreover, many lack any teaching experience themselves.

Conversely, our researchers have extensive knowledge of the subjects they teach. However, this does not necessarily imply that they know how to teach that subject effectively - referred to as *pedagogical content knowledge* [6].

In this context, I position myself at the intersection of these domains. I can be considered an educationalist - but conversant with educational literature - with both content knowledge and pedagogical content knowledge. I understand how lab courses can be designed and why certain measures approaches work or do not work. I focus on teaching labs, but also teach introductory Python and I am becoming increasingly engaged in the pedagogy of teaching quantum mechanics (see next chapter). Many other physics related topics I have thought (albeit at a different level) and I thus know the struggles students are coping with. Deep knowledge of both education and physics allows me to interpret student responses in a content-informed way, e.g., when asking a student *"why is an electron not a particle (or equally valid, not a wave)?"*, I can assess not only the correctness of the answer but also what it reveals about their understanding of the dual nature of the electron. (physics & education)

This combination of expertise enables me to support colleagues in translating general educational principles into discipline-specific practice. As a result, I am frequently consulted by researchers on educational questions. Examples include Gary Steele on the renewal of the Electronic Instrumentation course, Roel Smit on curriculum coherence, Margreet Docter on implementing a Just-In-Time-Teaching approach [7] for the NB Electronics course, and Erik van der Kolk on the renewal of the second-year lab course, and Rolf Hut in e.g. the swift transition to remote teaching with Covid, among others. Importantly, this role extends beyond our faculty's educational programs: I am also consulted by colleagues in Aerospace Engineering, Mechanical Engineering, and Civil Engineering.

Notably, this role also contributes to the broader educational program: several changes implemented in the First Year Physics Lab Course (FYPLC) have been adopted in other courses, particularly Design Engineering for Physicists (DEF) and the second-year lab course. In this way, my position at the intersection of content and pedagogy enables me to contribute not only to individual courses, but also to the coherence and development of the curriculum as a whole. physics

2.4 Vision statement on education

Philosophy: My philosophy in and on education is that I want students to develop (in a structured way) a sense of what (scientific) quality entails. For teaching in the first year physics lab course, this means that I want students to learn how to make well-considered choices throughout the research process. There is no single method in science, and there are no set procedures that always lead to the best result. The question at the core is *What decision leads to the best result in the given circumstances?* Hence, rather than telling students what to do - or let them following a recipe - they should consider the quality of their choices and actions throughout the process and, ultimately, be able to argue why their decision is defensible in light of the constraints at that specific moment in time.[2]

In the teacher-education program I use a similar approach: there are so many ways to engage pupils in learning physics, I hardly can cover all of them. However, we might enable pre-service teachers to develop a sense of what quality teaching is, recognize it and therefore understand whether to adopt and adapt a certain teaching strategy or not.

This philosophy sets me as a teacher not in front of the class but in the middle of the classroom, coaching and guiding students in their learning process. I see my role as that of a facilitator who creates an environment (and materials) where students can explore, experiment, and learn from their experiences. This is not always successful as some students are more acquainted with the idea *"tell me what to do, and I will do that (for you)"*. However, in the end they all seem to benefit from my approach.

Technology: I like exploring new ways of teaching, especially with the use of technology. I explored and implemented the use of interactive textbooks (see impact section), and the use of ChatGPT. The latter might especially help students in writing - where I see a lot of potential but also want to be cautious that the AI is not taking over their thinking (and thus their learning process). Again, I want students to be able to develop a sense of what quality is, recognize it and therefore understand whether to adopt and adapt a text produced by genAI. Moreover, I have some caution as there is so much fun in finding things out [8].

Future: I aspire to continue my role in both the educational program of Applied Physics and the teacher-education program. I enjoy teaching physics, in particular teaching scientific inquiry. At the same time, I am increasingly aware of the societal urgency of the teacher shortage. I believe that universities have an important responsibility in addressing this challenge. Universities should not only be educating future engineers and scientists, but also actively stimulating and supporting students who may consider a career in education. Besides, I really like inspiring future teachers and enjoy their creative ways of engaging pupils in physics.

this would benefit the [illegible] as well as with more and better prepared teachers pupils can be better prepared to ...

In a dual role as physics educator in the Applied Physics program and physics teacher educator in the teacher-education program, I believe I can contribute optimally to both programs. Through my involvement in the bachelor program, I reach approximately 200 students per year at an early stage in their academic development. This allows me to make a larger number of students aware that the teacher-education program exists and to inspire them to at least consider a career in education.

Moreover, I see strong potential for cross-pollination between the two programs. My involvement in teacher education allows me to bring current insights from physics didactics into the Applied Physics curriculum. Conversely, developments in university-level physics education (such as innovations in laboratory work, inquiry-based learning, and the use of open and digital tools) can enrich the teacher-education program. In this way, the dual role strengthens both programs and contributes to a more coherent and future-oriented educational ecosystem within the university.

2.5 Reflection

I believe that the above demonstrates that I am not only able to teach physics at different educational levels but also that I possess the skills and experience to innovate educational programs in a sustained and systematic manner, both at the level of individual courses and at the level of the curriculum. Furthermore, my work extends beyond teaching and local educational practice. I actively contribute to the development of education by disseminating insights, materials, and results through professional and academic journals, as well as through national and international conferences (see Appendix A). In doing so, I connect educational practice with scholarly reflection and contribute to the broader community of physics and physics education educators. Taken together, these activities reflect my readiness to take on greater responsibility and leadership in education, in line with the role and expectations of an associate professor with an emphasis on education.

both in physics as in teacher education

[2] I will come back to this in my research section.

3. Research

3.1 Research line into the integration of argumentation in science & engineering education

Argumentation is at the core of scientific practice. It is the means by which scientists construct, justify, and communicate knowledge, ensuring that claims are rigorously tested and defended. From formulating hypotheses to interpreting data, from engaging in peer review to debating theoretical frameworks, argumentation permeates every aspect of scientific inquiry. Without it, science would not function as a self-correcting, evidence-based discipline.

The ~~S. E.~~ Toulmin [9] model of argumentation has been widely used to describe the structure of arguments. It emphasizes key components such as claims, warrants, and backings, which form the foundation of a cogent scientific argument. However, scientific argumentation is more than just a method of presenting findings in a scientific journal in which all elements of the Toulmin argumentation are recognizable. Argumentation is an essential cognitive process in conducting research, the crucial element in the entire scientific endeavour. A scientific investigation does not merely seek to gather data; it aims to construct a compelling and well-supported claim based on that data. This requires reasoning systematically, evaluating alternative explanations, and providing justification for methodological choices.

Scientific argumentation thus plays a fundamental role in science, especially in inquiry and in engineering design. In the process of conducting research and designing technological solution, scientists and engineers do not passively record observations or build prototypes from scratch; they actively engage in constructing an argument that supports their conclusions/solution. This involves critically assessing evidence, scrutinizing methodologies, weighing pros and cons for different solutions to each problem they have to tackle or decision they have to make, and anticipating potential counterarguments.

In education, fostering argumentation skills is crucial for developing students' ability to engage in scientific reasoning. Research shows that many students view scientific inquiry as a mechanical process of following procedures rather than as a means of constructing and defending knowledge [10]. By integrating argumentation into science curricula, educators can shift students' perspectives, helping them understand that science is about producing the most reliable and defensible answers to research questions rather than merely completing experiments.

However, many studies relating to argumentation in science education focus primarily on the structure of the argument (e.g. identifying claims, warrants and backings) rather than looking at the content itself - the scientific cogency of the argument: what arguments are used as backing, what is the quality of the evidence, and how do students decide what data to collect and how to interpret ~~it~~. these However, if we want to teach students (at whatever level) how to setup a rigorous scientific inquiry or design a durable and long-lasting solution to a technical problem, we need to go beyond the structure of the argument. We need to understand how students make decisions throughout the research and design process, what knowledge they use to justify their choices, and how they can be supported in developing a more sophisticated understanding of scientific practices.

My PhD research focused on integrating argumentation into scientific inquiry education at the secondary school level [2]. Building on this work, my current and future research aims to further explore the integration of argumentation in both scientific inquiry and engineering design education. The goal is to develop theoretical frameworks and practical teaching approaches that help students not only understand the structure of scientific arguments but also engage deeply with the content and reasoning processes that underpin scientific research and engineering design. To further elaborate on what this entails, I provide two sections below that outline the research directions in scientific inquiry and engineering design education, respectively.

Figure 3.3: The Toulmin argumentation model integrated with the PACKS model, linking argumentation elements to types of knowledge required for scientific research.

3.1.1 Integration of argumentation and scientific inquiry

Enabling students to engage in ~~basic~~ scientific research is an important component of the curriculum in secondary and higher education [REF]. However, literature indicates that meaningfully implementing this (developing research skills ~~and~~ knowledge in students) is only minimally successful [REF]. In my PhD research, we have taken steps to address this issue [2]. We have demonstrated that argumentation is an indispensable but underemphasized aspect of education in scientific research. We also demonstrated that a focus on argumentation (viewing research as the construction of a scientifically convincing claim) leads students to develop a need for research knowledge, as they recognize that only the best answer to the research question suffices [11].

By defining 'research' as *the construction of the most convincing possible answer to a given question*, it is no longer sufficient for a student to merely apply the correct theory and perform methodologically sound procedures. The development of such routines remains important, but the didactics must also ensure that the student is continuously aware of the nature and purpose of the research activity. This means that the student makes decisions and choices that are rationally justifiable as they lead to optimally valid and reliable research outcomes within the given contextual constraints. This perspective on a 'research learning trajectory' requires the development of didactics that span from students' first encounter with scientific research to full-fledged independent physical research.

At the end of ~~the~~ dissertation, an extensive theoretical model is developed (see Figure 1). This model appears to be useful for further integrating argumentation and research, it links the Toulmin argumentation model [9] with the Procedural and Conceptual Knowledge in Science (PACKS) model [12]. While the Toulmin model describes the elements of an argumentation structure, the PACKS model outlines the different types of knowledge required to define the content of those elements. This theoretical model needs further elaboration and substantiation. This will be achieved through a theoretical paper with an extensive literature review, in which:

1. the necessity of integrating argumentation and research will be further argued,
2. the theoretical model will be expanded, and
3. its practical implementation (along with the associated implications for education) will be developed.

Various follow-up studies have been proposed in the dissertation that require further elaboration:

- If research knowledge is developed in physics education, to what extent is this knowledge relevant in other subjects, and how can we facilitate this knowledge transfer?
- What knowledge do teachers actually need, and how can we optimally train them to teach research skills? In collaboration with TUD's teacher education program, particularly within master's research projects, various studies are and will be conducted to address these questions.

Funding for ongoing research is being sought through one of the NRO calls for long-term research.

3.1.2 Integration of argumentation in engineering design

Engineering Design (ED), the systematic and creative process of ~~devising~~ and constructing solutions for a specific problem, is a key component in STEM education [13], [14] and is integrated in the intended learning outcomes of all Dutch secondary science subjects [15]. ~~ED~~ is mainly meant to:

- develop conceptual knowledge, strengthening (theoretical) subject matter taught through direct instruction [16];
- develop design and research practices, learning how an iterative process of thinking and tinkering may yield a prototype that solves the problem and suits the needs [17], [18], [19].

Engaging secondary school students in ED can enhance their real-world problem solving skills [20], foster interest in engineering careers [21], and provide illustrative preparation of and lay foundational groundwork for tertiary engineering education. Although it plays a central role in science curricula [13], [22], [23] and has a (potential) function in education, the teaching and learning of ED is problematic in various ways and the development of pedagogical approaches for ED fails to match EDs importance [24]. Students – at all levels – often struggle with ED when prerequisite design knowledge is not identified and developed timely. They tend to focus exclusively on the first idea that comes to mind without considering alternatives – known as design fixation [25]. Once the design is fully developed and a prototype built, the design choices are justified in retrospect (post hoc), only because assessment procedures demand it. Proper ED, however, requires that conscious, deliberate choices are made throughout the design process in order to justify the design [26]. Instead, as an expert design teacher stated: “We seem to teach our students how to come up with rationales for their designs on the spot.”

One of the main problems is that ED is often taught by educators with limited design experience (pre-tertiary level) or by design experts (tertiary level) who frequently lack pedagogical training or possess only implicit understanding of the practices and knowledge they apply [24], [27]. This dichotomy hinders effective teaching, leading some teachers to skip ED altogether, or to present it as an undifferentiated whole, rather than addressing specific aspects before applying them in an integrated manner [16]. Hence, for teaching ED at all levels a theoretical framework is essential for elucidating the intricacies of engineering design, explicating the requisite knowledge, and optimizing instructional approaches. In the related field of teaching scientific inquiry, a sequenced, argumentation focused (knowing what, knowing how, and knowing why) approach yielded an enhanced understanding of scientific practice and encourages students to develop personal reasons for adhering to scientific criteria [2]. I see opportunities for extending this approach to ED pedagogy and ‘uncover’ the yet unexplored area of integrating argumentation and ED.

ED, although often seen as particularly a creative discipline, should be considered a knowledge based activity, where more pertinent knowledge leads to better solutions. As in scientific inquiry, I believe that it should be possible to create a procedural and conceptual knowledge in science (PACKS) model that distinguishes different knowledge types that inform the decisions taken in various stages [12]. Unravelling the knowledge types allows to teach these in (mere) isolation -reducing and focussing the cognitive demand- before integrating these. These knowledge types include the idea that only the best solution (in the given circumstances) suffices, and understandings of what characterizes solutions that engineers regard as ‘best’, what understandings contribute to that evaluation, and how those solutions can be produced. Without these understandings, students may regard ED as ‘find a solution to the design problem, any solution will do’ prevents students from adopting and adhering to the norms and standards designers apply in their work.

These knowledge types are not yet identified for ED and thus need an elaborate theoretical framework that elucidates the intricacies of engineering design. Using the central questions:

- What are the understandings required to successfully design, conduct, and evaluate an engineering design?
- What are the characteristics of a valid, reliable, sufficiently specific, and detailed assessment of students’ understanding in engineering design?

I aim to construct a theoretical framework, (PACKED-model) that links the various knowledge types to the stages of the design process. I aim to derive a tentative set of knowledge types using literature- and field studies. Using these knowledge types, we can derive so-called field-dependent elements [9], which are the standards for judging the soundness, validity, cogency or strength of arguments. To propose, validate and test an approach to derive the presence and attainment level for these understandings, I will adapt the modified Delphi approach [28] as used in C. Pols, P. Dekkers, and M. De Vries [29].

Subsequently, using the central question: *How does the theoretical framework and a focus on argumentation facilitate teachers to design ED teaching-materials?* I aim to involve secondary school pre-service teachers in our teacher-education program and in-service teachers in a professional learning community [30] to use the theoretical construct and derive design principles to design teaching materials and test these in their own practices. Using education design research [31], we can investigate the teacher's use of the framework and its facilitation in designing lessons. The developed materials will ultimately be arranged in a coherent teaching-learning sequence, where the need for integrating argumentation and ED is emphasized: each and every design decisions should be justifiable in light of producing the best available solution within certain practical constraints (finance, time, knowledge).

Again, a study of this nature and size requires long-term funding, which is being sought through one of the NRO calls for long-term research. However, ~~in the field of educational research, there are limited opportunities for funding that support the hiring of PhD students~~. Unlike traditional applied physics research, physics education research does not typically fall within the scope of major research grants that provide long-term funding for doctoral candidates. There are research grants for education, but these foremost are focussed on education in general - resulting in a very divers field of applicants and reviewers (not aware of the issue in a specific educational domain). Moreover, teaching 6–8 day parts per week, being heavily involved in the new building, and serving on various committees at both local and national levels have limited the time available to focus on research. Despite these constraints, I have actively pursued funding opportunities to continue the research as outlined above. I have shown that with limited resources, I still can continue doing research on physics and engineering education [32], [33].

3.2 Supervising

As a faculty member in Applied Physics with a research focus on physics education, my work bridges the gap between fundamental physics and the scholarship of teaching and learning. My research aims especially on enhancing physics education by integrating inquiry learning, argumentation, and conceptual understanding into instructional practices.

Given the structural realities of education research - the limited financial resources available, I now build research capacity through mentoring, collaboration, and embedded research. For instance, in addition to my own research endeavors, I (try to) play a role in supporting assistant/associate professors and colleagues who wish to investigate their own teaching practices. Moreover, several papers have been written with students on their projects done in the first year physics lab course, allowing them to gain some experience in the publishing process. Through educational grants, which often focus on curriculum innovation rather than dedicated research positions, I help guide and supervise projects that lead to meaningful educational advancements. These grants provide faculty members the opportunity to develop and implement educational innovations, and I contribute by offering expertise, mentoring, and collaborative support to ensure their success.

Recently Rutger Ockhorst has started his PhD research on 'characteristics of a quantum mechanics teaching activity fostering a *need to know* in students'. We ~~(me as a daily supervisor)~~ have been working towards this start since 2023, where funding was sought for his project. During this initial phase, we have done several smaller studies resulting in two professional papers (see Appendix B). In the next years he will focus on the development of a quantummechanics activity, most likely an experiment that is linked to the work and experiments done in QTech, that shows the particle and wave behavior of electrons. In both secondary and tertiary education, this dual nature of electrons are taken for granted - students seem hardly puzzled about it where physicists are. This project will investigate what the characteristics of such an activity are in order to be successfull, and whether it is feasible to transform an experiment done in QTech or QuantumNanoscience to an educational setting.

3.3 Review

I have been actively involved in review work for scientific journals (PRPER), professional journals (phys. ed. for which I am an IoP's trusted reviewer.), conference proposals and proceedings (GIREP), and Dutch research proposals (SoTL & NRO klein) & TUD OESF. Furthermore, for the NRO's Scholarship of Teaching and Learning, I lead a TUD review committee to support teacher who applied for the NRO grant "scholarship of teaching and learning (SoTL)".

3.4 Reflection

I believe that the above illustrates that I found a niche in research that is worthy and interesting to pursue. I want to dedicate the next years to further develop this research line, and to contribute to the field of physics education research - at both secondary and tertiary level. Although I have not had the opportunity to formally supervise PhD students yet, I am actively involved in mentoring and guiding researchers in this domain. Moreover, I am recognized and frequently consulted by colleagues and editors to review educational research proposals and manuscripts. Taken together, I am confident that I can make meaningful contribution to the research as well as serve our university by disseminating this knowledge to our students and colleagues.

ref ref

4. Valorization & Impact

My main focus in valorization is on open education, which I see as a crucial aspect of my role as an educator and researcher. I also believe that I have had greatest impact in this area. Open Education can be described as *the (re)designing and sharing of learning materials, teaching methods, and practices that are freely accessible to everyone.* Open pedagogy goes beyond merely making materials open access available; it also encompasses the open sharing of the associated didactic approaches and pedagogical strategies used to provide effective education. It trusts on an evidence-based approach to education where the design of materials is informed by research and best practices in teaching and learning.

My commitment to open pedagogy is reflected in my efforts to not only create open educational resources but also to openly share the pedagogical principles and methods that underpin these materials. This includes documenting best practices, sharing experiences, and fostering a culture of collaboration. I actively engage in these activities at local, national, and international levels.

4.1 Local

I

At our university I initiated and participated in various initiatives to promote open pedagogy. For example, I contributed several times to the *TU Delft Education Day*, *TNW education day* and several *events at the Teaching Lab*, e.g. Python4All. I also organized a TU Delft-wide day focused on Python for educators, emphasizing the use of open tools and methods in teaching. I was part of the daily *PRIMECH* board and continued fostering quality Engineering Education as member of the *PRIMECH advisory board.*

I

In the *committee for the establishment of the Revising Academic Position Criteria report* (resulting in The Academic Development Guide), I advocated for the recognition of the importance of open education in academic career paths[3]. Additionally, I participate in the *Open Interactive Learning Materials* working group within TU Delft, focused on developing and promoting open learning materials and pedagogical approaches. In this committee we define guidelines and strategies regarding open education (especially the publication of books and learning materials). Moreover, the committee is engaged in developing strategies, workshops and templates allowing educators to create open learning materials more easily. Recently I became part of the *Open Science Festival committee*, responsible for organizing the 2026 Open Science Festival at TU Delft.

beleid

I also contribute to COMBINE, a Comenius initiative aimed at co-creating open learning materials with students. In this project I focus on the tools that allow both teachers and students to create interactive materials using Jupyter Book technology. Related to this, I am involved in the development of Teachbooks. A TU Delft initiative easing the way how open textbooks are created. Involvement in these initiatives allow other to create shareable resources more easily, thereby promoting open pedagogy. With regards to the latter, the initiative was ~~awarded~~ granted the 2026 TU Delft Open Education Inter-Faculty Team award.

Together with Rob Mudde, I created the book Introducing Classical Mechanics & Special Relativity. This book is now used in the first year of the bachelor physics program at TU Delft, and is openly available for others to use and adapt. It has gained already attention at international level and shows how physics can be taught interactively, even when students are at home. JBOSS

I initiated and lead ~~an~~ the inter-faculty open science project. Together with Saullo Castro (AE), Gary Steele (AS), Rolf Hut (CiTG), Ilke Ercan (EEMCS) and others we use Jupyter Book to enable students and researchers to collaboratively develop interactive, version-controlled outputs that can be published simultaneously as a website and a PDF (as exemplified by this document). In doing so, the JBOSS-project contributes to a reproducible and transparent research workflow, lowering the threshold for making data and code openly available. Beyond individual projects, the ambition is to develop an

[3] I have been asked to contribute to the development of the Academic Development Guide for all UD's, UHD's and Professors at TU Delft, replacing the Performance (WP) criteria. I was asked by the TU Delft Recognition & Rewards Committee to act as an expert in focus groups and individual consultations – particularly for the result area Education and the Career Path with Emphasis on Education Handbook.

~~(it is a flagship (?) showcasing latest publishing features...~~

infrastructure - through templates, manuals, and integrations - that supports both educational and research publishing. In this way, JBOSS acts as a bridge between open education and open science, enabling shared workflows and fostering a culture of openness across teaching and research. By introducing early-career researchers (students) to these practices, we contribute to embedding open science principles in the next generation of scientists. Importantly, this is no longer merely a vision but a demonstrated workflow: although still in its infancy, it is already being used by students with ease.

(a) the title page (b) the cover page

Figure 4.4: An impression of the Typst Thesis Template that was developed in the JBOSS-project.

4.2 National

At national level, I have been actively involved in various ways to promote a culture of open pedagogy and teacher professionalization. On a regular basis I report on educational practices in the Nederlands Tijdschrift voor Natuurkunde and the NVOX, the professional journal of the Dutch Association for Education in the Natural Sciences, see Appendix C. I have organized and contributed to national conferences (landelijke practicumdag) and workshops focused on open education (NPulse / Surf), sharing insights and best practices with educators across the country. Next to that I am involved in teacher professionalization in various ways: I have organized trainings and workshops on Arduino in secondary physics education, python for teachers, and teaching physics inquiry. Moreover, I am in the board of the Werkgroep Natuurkunde Didactiek (WND), the Dutch Physics Education organization that organizes a yearly professional development conference for ∼600 physics teachers. Collaboratively with several physics teacher educators, I have published ~~several b~~ooks and chapters for secondary school physics teachers, see Appendix C. Based on my work, especially C. Pols, P. Dekkers, and M. De Vries [29], the Hogeschool Rotterdam, Hogeschool Utrecht and Utrecht University have redesigned their lab courses.

4.3 International

At international level I contribute with workshops and presentations at large conferences (GIREP, AAPT, CDIO), and through articles in AAPT's The Physics Teacher and IoP's Physics Education, see frequently JB

Appendix C. Where I first utilized the tools that were offered by JupyterBook, I now actively contribute to the ecosystem by developing workshop materials, LaTeX and Typst templates and starterkits further positioning TU Delft as a frontrunner in open, interactive scientific publishing. This is highlighted in being an invited speaker at the Linux Foundation funded JupyterBook workshop 2026 at the Paris Saclay University. I presented the efforts at TU Delft to stimulate open education using Jupyter Book. Moreover, I regular connect with the core developers and help to make design decisions by reflecting on the user perspective. Lastly, I have gained access to the JupyterBook documentation and help in writing the manual, see Appendix C for a list of contributions. Noteworthy, I have been asked to help out setting up learning materials using Jupyter Books at the universities of Nijmegen, Leiden and Twente so far.

Together with TU Delft colleague Peter Dekkers we published the interactive book Show the Physics as an open educational resource. This book is now used internationally to professionalize pre- and in-service physics teachers.

4.4 Reflection

My involvement in open education, particularly through the Jupyter Book ecosystem, contributes to the open sharing of high-quality educational resources, enabling a broad and diverse audience to engage with and benefit from these materials. Importantly, this openness does not only increase accessibility, but also enhances quality: sharing materials often leads to feedback and suggestions from a wider, and sometimes unexpected, community of educators.

By not only sharing resources, but also developing and documenting workflows and tools that support their creation, adoption, and adaptation, I contribute to a sustainable ecosystem in which educational materials can continuously evolve and improve. This approach facilitates reuse and lowers the threshold for others to engage in open pedagogy.

Operating at an international level, this work contributes to positioning TU Delft as an active participant in the global open education movement, and aligns with the university's mission on lifelong learning by making high-quality educational resources accessible beyond the boundaries of our institution.

As an associate professor, I would like to continue my efforts in fostering open pedagogy at local, national and international level. Several initiatives have already been initiated or thought of, e.g. a national meeting on the use of Jupyter Book in Open Science (together with colleague Georgios Varnavides). Moreover, the TU Delft funded open science project *Connecting and boosting open science and open education through JB2* has shown feasible and viable. I want to further lead the project and change the way how we disseminate students' thesis at TU Delft. Both the Comenius teacher fellowship and Comenius senior fellowship (subsequently), are grants I would like to pursue.

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5. Leadership

Research on (engineering) education, education innovation and teaching professionalization are not centrally organized at our university. These related areas are ~~still~~ scattered whilst a more central and interconnected approach to these three areas would likely lead to a more effective and impactful education and educational research, as well as a more efficient use of resources. In many instances external experts are asked to give a lecture or training where in-house expertise is available. It is important to note that our own experts are more familiar with the educational and research context of our university and can therefore better tailor their support.
In the next five years, I ~~will~~ work on connecting the various actors and initiatives in these three areas, and I would like to work on positioning the Science & Engineering Education group in a more central role in education and educational research at our university. This is daunting and challenging, below I elaborate on the problems in the three areas

Various **educational research** projects and initiatives exist at our university. With the IDEE program, research on engineering education is linked to the Teaching Academy. However, Phd candidates and postdocs are hosted at the faculty of the main projectlead. The projectlead is not necessarily conversant with educational research and the young researcher is not housed in a educational research group. Moreover, some 'small' educational research projects are done at IDE, TBM, Architecture and AE, all contributing to the fragmented way how educational research is conducted at our university. A consequence of this fragmented approach is that the available resources (educational research knowledge, books, journal clubs) are not effectively shared - and therewith experiments are repeated with known outcomes, innovations are implemented incorrectly or with delays. Moreover, as educational research is different in many ways from engineering research, the PhD candidates and postdocs are not optimally supported in their research and early career development. exam...

Teaching professionalization (both UTQ and SUTQ) are **hosted** by the teaching learning service and housed in the teaching lab. As yet, there is no connection to the teacher education program or with the Science & Engineering Education ~~department~~ section. Again, the disconnect between the teaching professionalization and the teacher education program is a missed opportunity: modules, reflection forms and assessments are not shared, but also experiences and expertise are not exchanged. Hence, there is a duplication of efforts and missed opportunities for collaboration and educational growth. Uni

Teaching innovations are mostly left to individuals, especially young academics or junior teachers. There is no central support for teachers to innovate their education, and there is no central approach to sharing and scaling up successful innovations. This is a missed opportunity as well as it leads to a lot of duplicated efforts and lesser impact on teaching quality that could be achieved. Moreover, writing for teaching grants like the comenius teacher grant, or the scholarly of teaching are hardly supported and therefore not very common granted. when a more central

[illegible] that we should value and approach education in a similar way as we do our engineering research: collaboratively, well designed, interconnected, excellent, and thorough. A step forward in reaching this goal is to better align our efforts in teaching and innovating education and educational research. The Teaching Academy, teaching support, the Science & Engineering Education group and the (S)UTQ could be more closely connected and cooperating together in a collaborative effort to make education at our university more effective and impactful. limitations so far..

In my position as assistant professor I have been able to make personal connection with the various actors - I know many people from the UTQ, SUTQ, teaching support and the support team from the Teaching Academy. I also know many teachers from different faculties interested or doing educational research and have worked with them on a regular basis. As an associate professor I would like to take a more leading role in connecting the various actors and initiatives, for example by organizing regular meetings, workshops and joint projects - e.g. continuing educational research beyond the IDEE program, and defining a clear strategy on professionalization, innovation and research. My vision is to position the Science & Engineering Education group in a more central role in education and educational research at our university as they have a long-standing expertise in education and educational research. However, the focus has been mostly on secondary school education and only in recent years

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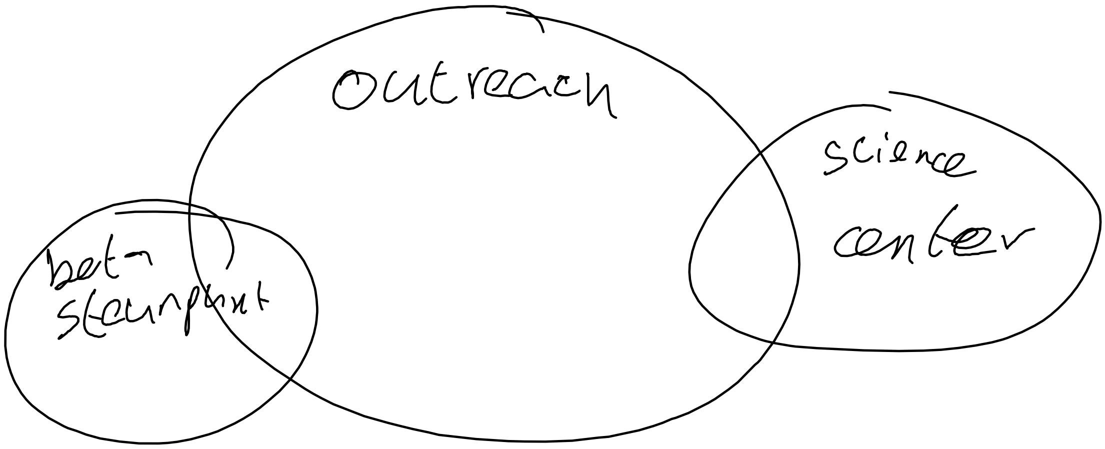
needs more thinking

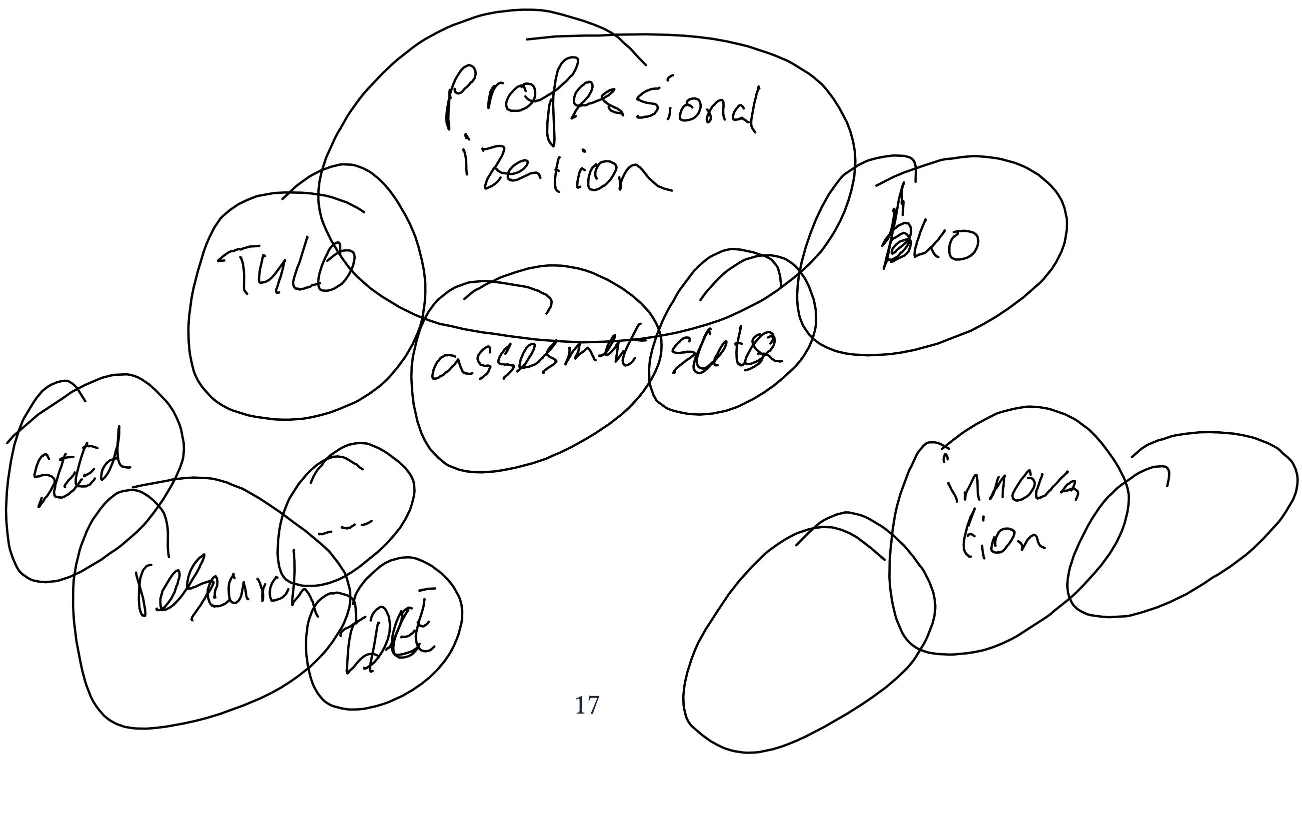
on higher education. Expanding the focus to higher education is a natural step as e.g. design education is already part of the research portfolio.

A more central approach to educational research would allow us to better support the PhD candidates and postdocs in their research, would increase the visibility of the group and the teacher education program (ultimately leading to more secondary school teachers) internally, increase the research profile in engineering education and thereby also increase the quality of education, increase the network leading to e.g. collaboration ,centralization (of educational initiatives), and invited speakers.

This is not a small task, but I believe that it is necessary to achieve our goals in education and educational research. Moreover, I believe that it is a task that fits well with my skills and interests, and that I can make a significant contribution to. The latter already demonstrated by the involvement of various cross faculty education initiatives.

Maar hoe dan, geef betere perspectief





6. Final remarks

never considered my self in being exceptional

To my dear colleagues,

The last seven years at Delft University of Technology have been a great experience to me. At the start of this document I stated that I pursued a PhD. as I had the feeling I had not reached my full potential. In my recent time at TU Delft, I have seen that I can grow even further and that it is possible to reach even higher levels of performance - not only by being smart but also by putting effort in it. I think such a growth mindset can set a great example for our students who often struggle in our educational program: I wasn't the brightest student in the study but by setting goals and working towards these, you can achieve great things.

I have seen a change at the TU Delft that is important here to mention: the university seems to value and reward education more and more. I am a little proud of being part of that change, not only by being a bystander who happens to have a focus on education, but by being an activist who has been actively involved in shaping the HR document, advocating open education, and promoting and fostering higher-quality education at our university. I look forward to the coming years in which I will continue the effort to approach education in a similar way as most of us approach research: by setting high standard goals, by fully dedicating ourselves to achieving these goals, and by doing this all in a collaborative way.

I am looking forward to hearing your feedback about this vision statement, and to the years ahead!

Best regards,

Freek Pols

7. Appendices

The appendices provide additional information on my academic activities categorized by the three different sections in this document.

7.1 Education

Below is a selection of relevant elements supporting the ... andere term

7.1.1 Courses

Course ~~code~~	~~Course name~~	Year	Role	Description
TN13010	Inleidend practicum 1	25 - now	Coordinator	A redesign of the FYPLC complying with the renewed physics curriculum. It includes now a 2 ECTS project on mechanics.
TN13015	Inleidend practium 2	25 - now	Coordinator	A redesign of the FYPLC complying with the renewed physics curriculum. It includes now a 2 ECTS project on thermodynamics.
TN1405	Inleidend practicum	19-25	Coordinator	The first year physics lab course (6ECTS ∼240 students) aims at introducing students to the basics of experimental physics.
TN2985	Introduction to experiments in physics	19 - 25	Coordinator	This course (2ECTS ∼40 students) introduces students how opted for the minor in physics to the basics of experimental physics.
WM0318TN	Wetenschap en argumentatieleer	22-23	Tutor	This course introduces the students to the philosophy of science (Nature of Science). In various sessions they discuss their standpoints regarding statements that are introduced in articles. They thereby explore their own views on the nature of science.
SL4301	Implementatie van onderwijs	23 - now	Guest lecturer	In this course students are introduced to relevant ideas and materials that they can use in their own teaching practices. I introduce them to the use of escaperooms and the use of open online interactive books.
SL4220	Vakdidactiek basis	22 - now	Guest lecturer	In this lecture I teach the basics of the pedagogy of practical work.

SL4320	Vakdidactiek verdiepend	22 - now	Guest lecturer	In this lecture I introduce the students to the use of four different ICT tools which can be used in the classroom.

Table 7.1: Courses taught

7.1.2 Dissemination of course innovations

This is a selection of the outputs related to the innovation of the lab courses.

- C. Pols [34]: *A Physics Lab Course in Times of COVID-19.*
- C. Pols and P. Dekkers [33]: *Redesigning a first year physics lab course on the basis of the procedural and conceptual knowledge in science model.*
- F. Pols, L. Duynkerke, J. Van Arragon, K. Van Prooijen, L. Van Der Goot, and B. Bera [35]: *Students' report on an open inquiry.*
- F. Pols [36]: *One setup for many experiments: enabling versatile student-led investigations.*
- F. Pols [37]: *Elements of proper conclusion.*
- C. Pols, H. Lewandowski, P. Logman, and F. Bradbury [38] *Comparison of practicum courses across four universities*
- F. Pols [39]: *The Vitruvian man: An introduction to measurement and data analysis.*

7.1.3 Supervised TA's beyond regular course duties

These TA's were engaged in developing content, making and discussing exams, helped develop software for the course and so on.

- Josh Sleijfer
- Hugo de Groot
- Hanna den Hartog
- Luka Vinck
- Laurent de Geus
- Timo Gort
- Suzanne Schuurman
- Luuk Fröling
- Maciej Topyla
- Dion Hoeksema
- Alek Heuvel
- Folkert Straus

7.2 Research

A selection of relevant research outputs supporting ...

7.2.1 Scientific publications

A detailed list of scientific publications with abstracts is available here.

- C. Pols and P. Dekkers [40] *Open Online Interactive Textbooks in Engineering Education: The Delft University of Technology Case.*
- J. Bickel, J. van der Veen, J. Beyers, and C. Pols [41] *Comparing and contrasting students' and faculty perceptions and utilization of GenAI.*
- L. G. de Putter-Smits, C. Pols, P. Dekkers, P. Runhaar, M. Timmer, and J. T. Van der Veen [32] *Exploring the role of generative AI in science teacher education programs: a qualitative study.*
- C. Pols and P. Dekkers [33] *Redesigning a first year physics lab course on the basis of the procedural and conceptual knowledge in science model.*
- C. Pols, P. Dekkers, and M. de Vries [11] *Integrating argumentation in physics inquiry: A design and evaluation study*
- C. Pols, P. Dekkers, and M. De Vries [42] *Would you dare to jump? Fostering a scientific approach to research in secondary physics education.*
- C. Pols, P. Dekkers, and M. De Vries [29] *Defining and assessing understandings of evidence with the assessment rubric for physics inquiry: Towards integration of argumentation and inquiry.*
- C. Pols, P. Dekkers, and M. De Vries [43] *What do they know? Investigating students' ability to analyse experimental data in secondary physics education.*
- F. Pols, P. Dekkers, and M. de Vries [44] *Introducing argumentation in inquiry—a combination of five exemplary activities*

7.2.2 Publications with colleagues

A selection of papers co-authored with colleagues on education.

- W. Bouwman, X. Feng, H. Geertsema, R. Mudde, and C. Pols [45] *Coherent teaching of mechanics with a theme park physics project*
- M. Docter and C. Pols [46] *It's all about the BASICs: A framework for a minds-on electronic course*
- C. Pols and T. Idema [47] *Eenvoudig online boeken maken met Jupyter Book*
- F. Pols and R. Hut [48]*Maker education in the applied physics bachelor programme at Delft University of Technology*
- R. Hut, C. Pols, and D. Verschuur [49] *Teaching a hands-on course during corona lockdown: from problems to opportunities*
- F. Bradbury and C. Pols [50] *A pandemic-resilient open-inquiry physical science lab course which leverages the Maker movement.*

7.2.3 Publications with Rutger Ockhorst

- R. Ockhorst, L. Koopman, and F. Pols [51] *A hands-on activity to introduce the structure of NV-center quantum bits in diamond*
- R. Ockhorst and F. Pols [52] *Development of the mental models of wave and particle as basis for wave-particle duality*

7.2.4 Publications with students

A selection of papers co-authored with students.

- F. Pols, L. Duynkerke, J. Van Arragon, K. Van Prooijen, L. Van Der Goot, and B. Bera [35] *Students' report on an open inquiry*
- F. Pols, N. Zwinkels, and S. Bulcke [53] *De radioactiviteit van fruit*
- C. Pols, R. Hut, L. Oosterlaan, F. Collenteur, and M. Braak v. [54] *Bouw een demoproef*
- F. Pols [36] *One setup for many experiments: enabling versatile student-led investigations*
- C. Pols, X. Perk, and T. A [55] *Developing inquiry experience through curiosity-driven open inquiry*

7.2.5 Master thesis supervision

20XX ~~Master Thesis~~ Diana ?

2024 ~~Master Thesis~~ Farah Kidwai: De SGO met perspectiefvragen - Een ontwerponderzoek naar het bevorderen van micro-macro denken in de scheikunde begripspracticumles

2025 ~~Master Thesis~~ Marcel Claus

2025 ~~Master Thesis~~ Mila Dobrovinski

7.3 Valorization

7.3.1 Educational papers (international)

A detailed list of educational publications with abstracts is available here.

- F. Pols [39] *The Vitruvian man: An introduction to measurement and data analysis*
- F. Pols [5] *The scientific graphic organizer for lab work*
- F. Pols and P. Diepenbroek [56] *Collaborative data collection: shifting focus on meaning making during practical work*
- F. Pols [57] *What's inside the pink box? A nature of science activity for teachers and students*
- F. Pols [58] *The sound of music: determining Young's modulus using a guitar string*
- C. Pols [59] *Real or fake? What can students learn from debunking Hollywood physics?*

7.3.2 Books & Chapters

- C. Pols and A. Mooldijk [60] *Leerstofdomeinen: Technische automatisering.*
- [61] *Leren onderzoeken*
- [62] *Practicumdidactiek*
- E. van den Berg and C. Pols [63] *Pocketdemo's*
- C. Pols and P. Dekkers, Eds. [64] *ShowthePhysics*
- I. Frederik *et al.* [65] *ShowdeFysica 2*
- I. Frederik *et al.* [66] *ShowdeFysica 3*
- R. Mudde, B. Rieger, and C. Pols [67] *Introduction to mechanics and special relativity*

7.3.3 Educational papers (national)

A detailed list of Dutch publications with abstracts is available here.

- C. Pols, P. Dekkers, and E. van den Berg [68] *Practicum, wat hebben we (ervan) geleerd?*
- C. Pols and T. Idema [47] *Eenvoudig online boeken maken met Jupyter Book*
- C. Pols [69] *Leerlingen leren onderzoeken*
- C. Pols and A. Mooldijk [70] *Melk in je koffie*
- C. Pols [71] *De natuurkunde van Spiderman*
- C. Pols [72] *Proeven (v/a)an vroeger*
- C. Pols and T. Levert [73] *Eerstejaarspracticum in tijden van COVID-19*
- C. Pols [74] *Practica bij de valbeweging*
- C. Pols [75] *Kritisch? Ik?! Ontwikkeling van een kritische houding in een leerlijn onderzoeken*
- C. Pols [76] *Practicum, net even anders*
- C. Pols [4] *De Scientific Graphic Organizer*
- C. Pols [77] *De mens van Vetruvius*
- C. Pols [78] *Leren onderzoeken: een praktische aanpak in klas 4*
- C. Pols [69] *Leerlingen leren onderzoeken*
- C. pols [79] *De hellingproef: Een practicum over het ontbinden van krachten in componenten*
- C. Pols [80] *Beweging in stop-motion*
- C. Pols [81] *Snaartheorie in de praktijk*
- C. Pols [82] *Technisch ontwerpen in 4V*
- C. Pols and J. Nelk [83] *Een leerlingpracticum quantummechanica voor minder dan € 30,-*
- C. Pols [84] *Een foto of grafiek zegt meer dan 1000 woorden*
- C. Pols [85] *Een versnelling langs een helling*
- C. Pols [86] *Een dubbele proef*
- C. Pols [87] *Competitie bootje vouwen*
- C. Pols [88] *Schoolboeken, toen en nu*
- C. Pols [89] *De eindtoets voor het basisonderwijs*
- C. Pols [90] *Vier keer schieten voor weinig geld*

7.3.4 Software & Educational tools

Jboss

- Teachbooks

A platform with an extensive start template allowing teachers to easily create their own educational resources.

- plain_typst_template

A minimal Typst book template for Jupyter Book users, allowing users to create a high-quality PDF version of their Jupyter Book content using Typst.

- typst_thesis_template

A template for master thesis using Typst with Jupyter Book.

- latex_thesis_template

A template for master thesis using LaTeX with Jupyter Book.

- Starterkit Tags

A starterkit which provides students a headstart in writing and organizing their thesis project.

- Myst plugins

This repro where plugins that extend the features of Jupyter Book was initiated by me. Several plugins were developed. Notably the last_updated_page, a plugin that shows when the page was last updated, is well appreciated and used in several projects.

- Jupyter Book

I contribute to the documentation of the Jupyter Book project. I created the gallery, made several edits and wrote the section on gitlab and gitlab pages.

JB
- Tutorial

Together with Jim Madge (the turing way) and Robert Lanzafame we created a Jupyter Book workshop which can be used for training purposes and is an easy entrance to Jupyter Book projects.

7.3.5 Grants & Applications

volgorde

Year	Title	Call	Main applicant	Accepted
2014	**The interactive whiteboard as intervention during practical work in physics class to increase students learning**	NRO: Promotiebeurs voor leraren	V	V
2023	**Perspective Agility for Navigating Diversity, Collaboration and Conflict in order to advance Equity and Inclusion**	NRO: Comenius Senior fellow	X	V
2023	A modular programming book for engineering students	TUD Open Education Stimulation Fund	X	X
2023	**Opening Up Classroom Demonstration Experiments**	TUD Open Education Stimulation Fund	X	V
2023	**The road to scientific inquiry**	TUD Education Fellowship	V	V
2023	To ChatGPT, or not to ChatGPT: Effectief gebruik van Artificial Intelligence in het onderwijs	NRO: Comenius teaching fellowship	V	X

Year	Title	Call	Main applicant	Accepted
2024	Integration of Argumentation and Engineering Design	NRO: edVenturous Questions	V	X
2024	Towards a coherent curriculum: Guiding curriculum transformation through educational design theory	NRO: Hoger onderwijs van de toekomst	X	X
2024	From Closed to Open: Bottom-up Open Educational Resources with and for pre-service teachers	NRO: Comenius teaching fellowship	V	X
2024	**Researching the renewed AP curriculum**	TUD Education Fellowship	X	V
2025	The Scientific Graphic Organizer as aid for practical work in science subjects	NRO: Pareltjes uit de praktijk	X	X
2025	gAI in secondary education – designing exemplary activities to develop student and teacher competences	NRO: Kennis voor de toekomst, Klein	X	X
2025	Integration of Argumentation and Engineering Design	NRO: Kennis voor de toekomst, Middel	V	X
2025	Building Thinking Classrooms in Wiskunde-, Natuurkunde- en Scheikundeonderwijs	NRO: Kennis voor de toekomst, Middel	X	X
2025	Student-Centred Learning with AI Tutors in Project Work	TUD: AI-Augmented Engineering Education	V	X
2025	**Jupyter Books for education & open publishing: Next Level**	TUD Open Science Fund	X	V
2025	Doe de Fysica	Betadecanen	V	X
2025	**An affordable and adaptive educational escaperoom**	TUD Open Hardware Fund	V	V
2025	**Connecting and boosting open science and open education through JupyterBook**	TUD Open Science Fund	V	V
2026	Connecting open science and open education through digital interactive, executable narratives	NWO	V	?

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